

Question 1: What is the Difference Between Descriptive Statistics and Inferential Statistics? Explain with Examples.

Introduction

Statistics is the branch of mathematics that deals with collecting, organizing, analyzing, interpreting, and presenting data.

Statistics is mainly divided into two types:

1. **Descriptive Statistics**
2. **Inferential Statistics**

The main difference is that **descriptive statistics summarizes the data that has been collected**, while **inferential statistics uses sample data to make conclusions or predictions about a larger population**.

1. Descriptive Statistics

Descriptive statistics is used to summarize and describe the main characteristics of a dataset.

It does not make predictions or generalizations beyond the data being analyzed.

Common Methods

- Mean
- Median

- Mode
- Range
- Variance
- Standard deviation
- Minimum and maximum
- Frequency tables
- Charts and graphs

Example

Suppose the marks of five students are:

60, 70, 75, 80, 90

The mean is:

$$\begin{aligned}\text{Mean} &= \\ & \frac{60+70+75+80+90}{5} \\ & = 75\end{aligned}$$

Therefore, we can say that the **average mark of these five students is 75**.

This is descriptive statistics because we are only summarizing the given five students.

Real-World Example

A company analyzes the monthly sales of its 100 employees and finds:

- Average sales = ₹50,000
- Highest sales = ₹90,000
- Lowest sales = ₹20,000
- Standard deviation = ₹12,000

These values describe the existing sales data.

2. Inferential Statistics

Inferential statistics is used to draw conclusions, make predictions, or make decisions about a **population based on sample data**.

Since it is usually difficult or expensive to collect data from an entire population, researchers study a sample and use statistical methods to infer information about the population.

Common Methods

- Estimation
- Confidence intervals
- Hypothesis testing
- Regression analysis
- Correlation
- t-test
- Chi-square test
- ANOVA

Example

Suppose a university has **10,000 students**.

Instead of asking every student about their average study time, researchers select a random sample of **500 students**.

If the sample shows that students study an average of **3.5 hours per day**, researchers may use inferential statistics to estimate the average study time of all 10,000 students.

Here:

Population → 10,000 students

Sample → 500 students

Sample result → 3.5 hours

Inference → Estimate for the entire population

This is inferential statistics because the sample is being used to make a conclusion about the larger population.

Difference Between Descriptive and Inferential Statistics

Basis	Descriptive Statistics	Inferential Statistics
Purpose	Summarizes and describes data	Draws conclusions about a population
Data Used	Usually the collected dataset	Usually a sample from a population
Prediction	Does not primarily predict	Can make predictions/estimates
Generalization	Does not generalize beyond the analyzed data	Generalizes from sample to population
Methods	Mean, median, mode, range, SD	Hypothesis testing, confidence intervals, regression
Example	Average marks of a class	Estimating average marks of an entire university
Main Question	"What does the data show?"	"What can we conclude about the population?"

Simple Example to Understand the Difference

Suppose a company surveys **100 customers**.

Descriptive Statistics

The company finds:

Average satisfaction=4.2/5

This describes the satisfaction of the **100 surveyed customers**.

Inferential Statistics

The company uses those 100 customers as a sample to estimate the satisfaction of **all 50,000 customers**.

This is inferential statistics.

Relationship Between the Two

Descriptive and inferential statistics are often used together:

Collect Data

↓

Descriptive Statistics

↓

Summarize the Sample

↓

Inferential Statistics

↓

Make Conclusions About Population

For example, a researcher may first calculate the **mean and standard deviation** of a sample and then use **hypothesis testing or confidence intervals** to draw conclusions about the population.

Key Point

The easiest way to remember the difference is:

Descriptive statistics describes what the data contains, whereas **inferential statistics** uses data from a sample to make conclusions about a larger population.

Conclusion

Descriptive statistics focuses on organizing, summarizing, and presenting the data that has been collected. It includes measures such as mean, median, mode, range, and standard deviation.

Inferential statistics goes one step further by using sample data to estimate, predict, or make conclusions about a larger population. It includes techniques such as hypothesis testing, confidence intervals, and regression.

Therefore:

Descriptive = Describe the data

Inferential = Infer conclusions about the population

Question 2: What is Sampling in Statistics? Explain the Differences Between Random and Stratified Sampling.

Introduction

In statistics, it is often difficult, expensive, or time-consuming to collect information from every member of a population. **Sampling** provides a practical way to study a smaller group and use the results to understand the larger population.

What is Sampling?

Sampling is the process of selecting a smaller group of individuals, called a **sample**, from a larger group, called the **population**, for the purpose of collecting and analyzing data.

Example

Suppose a college has **5,000 students** and we want to know their average satisfaction with college facilities.

Instead of asking all 5,000 students, we may select **500 students** and collect their responses.

Population

5,000 Students

↓

Sampling



Sample

500 Students



Collect & Analyze Data



Draw Conclusions

The **population** is the complete group, while the **sample** is the smaller group selected from it.

What is Random Sampling?

Random sampling, also called **simple random sampling**, is a method in which every member of the population has an **equal or known chance of being selected**.

The selection is made randomly, so the researcher does not intentionally choose specific individuals.

Example

A company has **1,000 employees** and wants to survey 100 employees.

The company assigns each employee an ID from 1 to 1,000 and randomly selects 100 IDs.

1,000 Employees

↓

Random Selection

↓

100 Employees

Every employee has an equal chance of being selected.

Advantages

- Simple to understand and implement.
- Reduces researcher selection bias.
- Provides a representative sample when properly conducted.
- Suitable when the population is relatively homogeneous.

Limitations

- Requires a complete list of the population.
- Can be difficult for very large or geographically distributed populations.
- Important subgroups may be underrepresented by chance.

What is Stratified Sampling?

Stratified sampling is a method in which the population is divided into smaller, relatively homogeneous groups called **strata** based on a relevant characteristic.

A sample is then selected from **each stratum**.

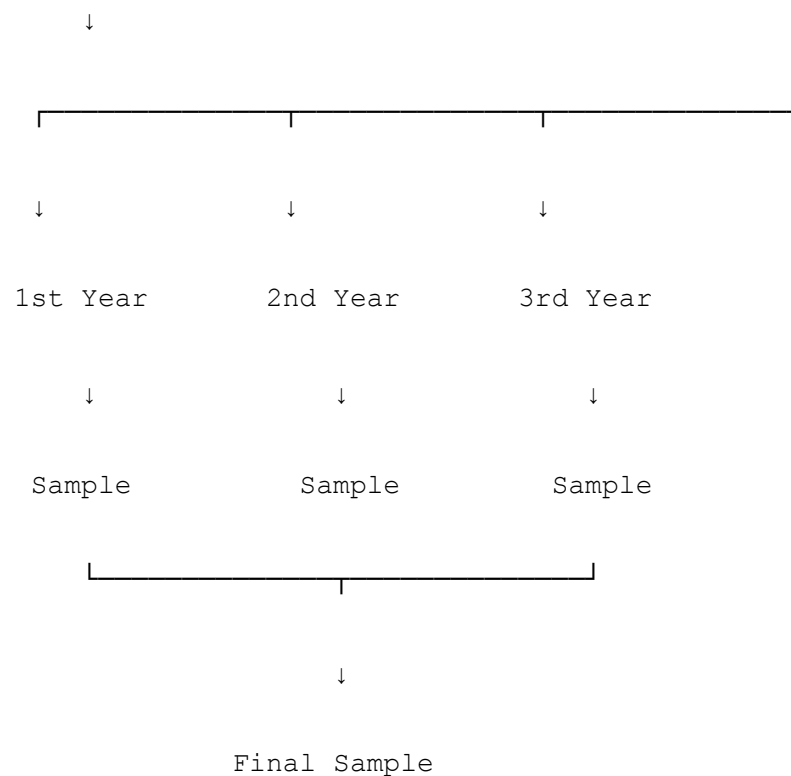
Strata may be based on:

- Age
- Gender
- Income group
- Department
- Education level
- Geographic region

Example

Suppose a college has 5,000 students:

Population



If the researcher wants every year to be represented, students can be divided into strata based on year of study and then randomly selected from each group.

For example:

Stratum	Population	Sample
1st Year	2,000	200
2nd Year	1,500	150
3rd Year	1,000	100
4th Year	500	50
Total	5,000	500

This is an example of **proportionate stratified sampling**, where the sample size from each group reflects its population size.

Difference Between Random and Stratified Sampling

Basis	Random Sampling	Stratified Sampling
Meaning	Selects individuals randomly from the entire population	Divides population into strata and samples from each
Population Division	No required division	Population is divided into groups
Selection	Random selection from the whole population	Random selection within each stratum

Main Goal	Give each member an equal/known chance of selection	Ensure important subgroups are represented
Representation	Some groups may be underrepresented by chance	Each selected stratum is represented
Complexity	Relatively simple	More complex
Best Used When	Population is relatively homogeneous	Population contains important distinct subgroups
Example	Randomly selecting 100 students from 5,000	Selecting students from each year of study

Simple Example

Suppose a company has:

- 600 male employees
- 400 female employees

and wants a sample of 100 employees.

Random Sampling

The company randomly selects 100 employees from all 1,000 employees.

There is no separate requirement to select from each group.

Stratified Sampling

The company divides employees into two strata:

Stratum 1 → Male Employees

Stratum 2 → Female Employees

Then it selects employees from both groups, for example proportionately:

- 60 males
- 40 females

This ensures that both groups are represented.

Advantages of Stratified Sampling

1. Ensures representation of important subgroups.
 2. Can provide more precise estimates when strata are well chosen.
 3. Useful when the population is heterogeneous.
 4. Allows separate analysis of different groups.
 5. Can reduce sampling variability compared with a simple random sample in suitable situations.
-

Conclusion

Sampling is the process of selecting a smaller group from a larger population to collect and analyze data.

- **Random sampling** selects individuals randomly from the entire population, giving each member an equal or known chance of selection.
- **Stratified sampling** first divides the population into meaningful groups or strata and then selects a sample from each group.

The key difference is:

Random Sampling → Random selection from the whole population

Stratified Sampling → Divide into groups → Randomly sample from each group

Therefore, **random sampling** is useful for simple and unbiased selection, while **stratified sampling** is particularly useful when ensuring representation of important subgroups is important.

Question 3: Define Mean, Median, and Mode. Explain Why These Measures of Central Tendency Are Important.

Introduction

Measures of central tendency are statistical measures used to identify the **central or typical value** of a dataset. They provide a single value that represents the overall distribution of the data.

The three most common measures of central tendency are:

1. **Mean**
 2. **Median**
 3. **Mode**
-

1. Mean

The **mean** is the arithmetic average of all the values in a dataset.

Formula

$$\text{Mean} = \frac{\text{Sum of all observations}}{\text{Number of observations}}$$

or

$$\frac{\sum x}{n}$$

where:

- $(\bar{x})$ = Mean
- $(\sum x)$ = Sum of all observations
- (n) = Number of observations

Example

Consider the following marks:

10, 20, 30, 40, 50

$$\begin{aligned}\text{Mean} &= \frac{10+20+30+40+50}{5} \\ \text{Mean} &= \frac{150}{5} \\ &= 30\end{aligned}$$

Therefore, the **mean is 30**.

Importance of Mean

- Uses all observations in the dataset.

- Easy to calculate and understand.
- Useful for comparing different datasets.
- Widely used in further statistical calculations.
- Suitable for numerical data that does not contain extreme outliers.

Limitation

The mean can be strongly affected by **extreme values (outliers)**.

For example:

10, 20, 30, 40, 500

The value 500 greatly increases the mean, even though most observations are much smaller.

2. Median

The **median** is the middle value of a dataset after arranging the observations in **ascending or descending order**.

Case 1: Odd Number of Observations

For:

10, 20, 30, 40, 50

The middle value is:

Median=30

Case 2: Even Number of Observations

For:

10, 20, 30, 40

There are two middle values: 20 and 30.

Therefore:

$$\begin{aligned}\text{Median} &= \\ &= \frac{20 + 30}{2} \\ &= 25\end{aligned}$$

Importance of Median

- Less affected by extreme values than the mean.
- Useful for skewed data.
- Suitable for income, salary, house prices, and other datasets containing outliers.
- Provides a good representation of the central position of ordered data.

Example

Consider salaries:

₹20,000, ₹25,000, ₹30,000, ₹35,000, ₹5,00,000

The very high salary of ₹5,00,000 affects the mean considerably.

The median remains:

$$\text{Median} = ₹30,000$$

Therefore, the median may better represent the typical salary in this example.

3. Mode

The **mode** is the value that occurs **most frequently** in a dataset.

Example

Consider:

2, 3, 3, 4, 5, 3, 6

The number **3** occurs most frequently.

Therefore:

Mode=3

Importance of Mode

- Easy to identify.
- Useful for categorical data.
- Shows the most common value or category.
- Not affected by extreme values.
- Useful in business and market analysis.

Example

A clothing store records the shirt sizes sold:

M, L, M, S, M, XL, L

The mode is:

Mode=M

This tells the store that **Medium (M)** is the most frequently purchased size.

Comparison of Mean, Median, and Mode

Measure	Definition	Main Advantage	Effect of Outliers
Mean	Arithmetic average	Uses all observations	Highly affected
Median	Middle value	Good for skewed data	Less affected
Mode	Most frequently occurring value	Useful for categorical data	Not affected

Why Are Measures of Central Tendency Important?

1. Summarize Large Datasets

A large dataset can contain thousands or millions of observations. Mean, median, or mode can summarize the data using a single representative value.

2. Easy Comparison

Central tendency allows us to compare different groups.

For example:

Class A Average = 72 marks

Class B Average = 65 marks

We can quickly see that Class A has a higher average performance.

3. Helps in Decision-Making

Businesses, governments, researchers, and organizations use central tendency to make decisions.

For example, a company can use the **median salary** to understand typical employee compensation.

4. Identifies the Typical Value

These measures help answer questions such as:

- What is the average income?
 - What is the typical house price?
 - What is the most common product size?
 - What is the middle exam score?
-

5. Useful for Further Statistical Analysis

The mean is particularly important in many statistical techniques, including:

- Variance
- Standard deviation

- Correlation
 - Regression
 - Hypothesis testing
-

When Should We Use Each Measure?

Situation	Best Measure
Normally distributed numerical data	Mean
Data contains extreme outliers	Median
Highly skewed data	Median
Categorical data	Mode
Finding most popular category	Mode
Comparing average numerical values	Mean

Simple Example

Consider the dataset:

5, 7, 7, 8, 9

Mean

$$\text{Mean} = \frac{5+7+7+8+9}{5}$$

$$\frac{5}{7} = 0.7142857142857143$$

Median

The middle value is:

$$\text{Median} = 7$$

Mode

The most frequent value is:

$$\text{Mode} = 7$$

Therefore:

$$\text{Mean} = 7.2$$

$$\text{Median} = 7$$

$$\text{Mode} = 7$$

Conclusion

Mean, median, and mode are the three major measures of central tendency used to represent the center or typical value of a dataset.

- **Mean** → Arithmetic average of the data.
- **Median** → Middle value after sorting the data.
- **Mode** → Most frequently occurring value.

They are important because they **summarize data, simplify comparisons, identify typical values, support decision-making, and provide a foundation for further statistical analysis.**

The easiest way to remember them is:

Mean = Average

Median = Middle

Mode = Most Frequent

Question 4: Explain Skewness and Kurtosis. What Does a Positive Skew Imply About the Data?

Introduction

Skewness and **kurtosis** are statistical measures used to describe the **shape and distribution of data.**

- **Skewness** measures the **asymmetry** of a distribution.
- **Kurtosis** describes the **tail heaviness and peakedness** of a distribution compared with a normal distribution.

These measures help us understand whether data is symmetric, tilted toward one side, or contains unusually heavy or light tails.

1. Skewness

Skewness measures the degree to which a probability distribution is **asymmetric** around its mean.

A perfectly symmetric distribution has approximately zero skewness.

There are three main types:

1. **Zero Skewness**
2. **Positive Skewness**
3. **Negative Skewness**

A. Zero Skewness

When a distribution is approximately symmetric:

$$\text{Skewness} \approx 0$$

Typically:

$$\text{Mean} \approx \text{Median} \approx \text{Mode}$$

Example

2, 3, 4, 5, 6

The data is approximately symmetric around the center.

B. Positive Skewness

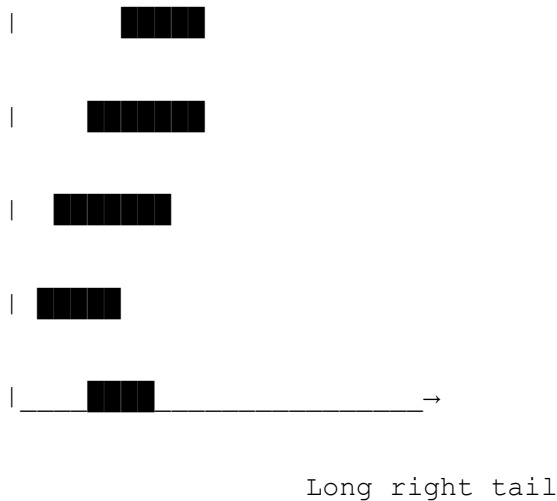
A distribution is **positively skewed** when it has a **longer tail toward the right**.

Frequency

|

|





In a positively skewed distribution, extreme high values pull the mean toward the right.

Usually:

Mean > Median > Mode

Example

Income data is often positively skewed because most people may have moderate incomes while a small number of people have very high incomes.

C. Negative Skewness

A distribution is **negatively skewed** when it has a **longer tail toward the left**.

Usually:

Mean < Median < Mode

Example

If most students score very high on an easy exam but a few students score very low, the distribution may be negatively skewed.

What Does Positive Skew Imply?

A **positive skew** implies that the distribution has a **long right tail**, meaning there are relatively more extreme or unusually high observations compared with a symmetric distribution.

It generally indicates:

- A few unusually high values are present.
- The mean is pulled upward.
- The **mean is usually greater than the median**.
- The distribution is not symmetric.
- The data may contain high-value outliers.

Example

Consider the following incomes:

₹20,000

₹22,000

₹25,000

₹27,000

₹30,000

₹2,00,000

Most incomes are between ₹20,000 and ₹30,000, but ₹2,00,000 is an unusually high value.

This creates a **long right tail**, resulting in positive skewness.

2. Kurtosis

Kurtosis describes the **tail behavior of a distribution**, particularly how much probability is concentrated in the tails relative to a normal distribution.

It is commonly used to identify whether data has relatively **heavy or light tails**.

There are three common categories:

1. **Mesokurtic**
 2. **Leptokurtic**
 3. **Platykurtic**
-

A. Mesokurtic

A **mesokurtic** distribution has kurtosis similar to a normal distribution.

For Pearson's kurtosis:

$$\text{Kurtosis}=3$$

For **excess kurtosis**:

$$\text{Excess Kurtosis}=0$$

The normal distribution is the standard example.

B. Leptokurtic

A **leptokurtic** distribution has **heavier tails** than a normal distribution.

It tends to have a greater tendency to produce extreme observations.

For excess kurtosis:

$$\text{Excess Kurtosis} > 0$$

C. Platykurtic

A **platykurtic** distribution has **lighter tails** than a normal distribution.

It generally has fewer extreme observations compared with a normal distribution.

For excess kurtosis:

$$\text{Excess Kurtosis} < 0$$

Skewness vs Kurtosis

Feature	Skewness	Kurtosis
Measures	Asymmetry	Tail heaviness
Main focus	Direction and degree of imbalance	Extreme-value/tail behavior
Value around normal distribution	0	0 excess kurtosis
Positive value	Right-skewed	Heavier tails

Negative value

Left-skewed

Lighter tails

Main use

Understand distribution's asymmetry

Understand tail behavior and extreme observations

Real-World Examples

Positive Skewness

Income distribution

A large number of people may have moderate incomes, while a small number have extremely high incomes.

Most observations → Moderate income

Few observations → Very high income

↓

Right tail

Kurtosis

Financial returns

A dataset with high kurtosis can have relatively heavy tails, meaning extreme gains or losses may occur more frequently than under a normal distribution.

Important Difference

It is useful to remember:

Skewness tells us about the direction of asymmetry, while kurtosis tells us about the behavior of the tails.

For example:

Skewness → Is the distribution tilted left or right?

Kurtosis → How heavy or light are the tails?

Summary

Concept	Meaning
Skewness	Measures asymmetry of a distribution
Positive Skew	Long right tail; high values pull the mean upward
Negative Skew	Long left tail; low values pull the mean downward
Zero Skew	Approximately symmetric distribution
Kurtosis	Describes tail heaviness relative to a normal distribution
Leptokurtic	Heavy tails
Mesokurtic	Similar to normal
Platykurtic	Light tails

Conclusion

Skewness and **kurtosis** help describe the shape of a dataset beyond measures such as mean and standard deviation.

Skewness measures asymmetry. A **positive skew** means the distribution has a **longer right tail**, usually because of some unusually high observations, and the mean is typically greater than the median.

Kurtosis focuses primarily on the **tail behavior and frequency of extreme observations** compared with a normal distribution.

Therefore:

Positive Skew→Long Right Tail

Skewness→Asymmetry

Kurtosis→Tail Heaviness

Question 5: Implement a Python Program to Compute the Mean, Median, and Mode

Given List

```
numbers = [12, 15, 12, 18, 19, 12, 20, 22, 19, 19, 24, 24, 24, 26, 28]
```

Python Program

```
import statistics
```

```
# Given list of numbers
```

```
numbers = [12, 15, 12, 18, 19, 12, 20, 22, 19, 19, 24, 24, 24, 26, 28]

# Calculate mean, median, and mode

mean = statistics.mean(numbers)

median = statistics.median(numbers)

mode = statistics.mode(numbers)

# Display the results

print("Numbers:", numbers)

print("Mean:", mean)

print("Median:", median)

print("Mode:", mode)
```

Output

```
Numbers: [12, 15, 12, 18, 19, 12, 20, 22, 19, 19, 24, 24, 24, 26, 28]
```

```
Mean: 20.8
```

```
Median: 20
```

```
Mode: 12
```

Explanation

1. Mean

The mean is calculated as:

$$\text{Mean} = \frac{\text{Sum of all values}}{\text{Number of values}}$$

There are **15 numbers**, and their sum is **312**.

$$\begin{aligned}\text{Mean} &= \\ & \frac{312}{15} \\ &= 20.8\end{aligned}$$

Therefore:

$$\text{Mean} = 20.8$$

2. Median

The numbers are already arranged in ascending order:

12, 12, 12, 15, 18, 19, 19, 19, 20, 22, 24, 24, 24, 26, 28

There are **15 values**, so the middle position is:

$$\begin{aligned} & \frac{15+1}{2} \\ &= 8\end{aligned}$$

The 8th value is **19**.

Therefore:

Median=19

3. Mode

The mode is the value that occurs most frequently.

Frequency:

- 12 → 3 times
- 19 → 3 times
- 24 → 3 times

Therefore, the dataset actually has **three modes**:

12, 19, 24

So this dataset is **multimodal**.

Important Correction to the Basic `statistics.mode()` Example

In recent Python versions, `statistics.mode(numbers)` returns the **first mode encountered**, which is 12. However, statistically, since **12, 19, and 24 each occur three times**, all three are modes.

To correctly display **all modes**, use `statistics.multimode()`:

```
import statistics
```

```
numbers = [12, 15, 12, 18, 19, 12, 20, 22, 19, 19, 24, 24, 24, 26, 28]
```

```
mean = statistics.mean(numbers)

median = statistics.median(numbers)

modes = statistics.multimode(numbers)


print("Mean:", mean)

print("Median:", median)

print("Modes:", modes)
```

Correct Output

Mean: 20.8

Median: 19

Modes: [12, 19, 24]

Functions Used

Function	Purpose
<code>statistics.mean()</code>	Calculates the arithmetic mean
<code>statistics.median()</code>	Calculates the median

`statistics.mode()` Returns a mode

`statistics.multimode()` Returns all modes

Final Answer

Mean=20.8

Median=19

Mode=12, 19, 24

Since three values occur with the highest frequency, the dataset has **three modes**.

Question 6: Compute Covariance and Correlation Coefficient Between Two Datasets

Given Data

```
list_x = [10, 20, 30, 40, 50]
```

```
list_y = [15, 25, 35, 45, 60]
```

We need to calculate:

1. **Covariance**
2. **Correlation coefficient**

Python Program

```
import numpy as np

# Given datasets

list_x = [10, 20, 30, 40, 50]

list_y = [15, 25, 35, 45, 60]

# Calculate covariance

covariance = np.cov(list_x, list_y)[0, 1]

# Calculate correlation coefficient

correlation = np.corrcoef(list_x, list_y)[0, 1]

# Display results

print("Covariance:", covariance)

print("Correlation Coefficient:", correlation)
```

Output

Covariance: 275.0

Correlation Coefficient: 0.949...

Therefore:

$$\text{Covariance}=275$$

$$\text{Correlation}\approx 0.949$$

1. Covariance

Covariance measures the direction in which two variables change together.

The sample covariance formula is:

$$\text{Cov}(X,Y)=$$

$$\sum$$

$$X$$

$$i$$

$$-$$

$$X$$

$$-$$

$$)($$

$$Y$$

$$i$$

$$-$$

$$Y$$

$$-$$

$$)$$

$$n-1$$

Step 1: Calculate the Means

For (X):

$$\begin{aligned} & \frac{X}{n} \\ &= \frac{10+20+30+40+50}{5} \\ &= 30 \end{aligned}$$

For (Y):

$$\begin{aligned} & \frac{Y}{n} \\ &= \frac{15+25+35+45+60}{5} \\ &= 36 \end{aligned}$$

Step 2: Calculate Deviations

X	Y	X – 30	Y – 36	Product
10	$\frac{1}{5}$	-20	-21	420
20	$\frac{2}{5}$	-10	-11	110
30	$\frac{3}{5}$	0	-1	0
40	$\frac{4}{5}$	10	9	90
50	$\frac{6}{0}$	20	24	480

Total

1100

Therefore:

$$\begin{aligned}\text{Cov}(X,Y) &= \\ & \frac{1100}{5-1} \\ \text{Cov}(X,Y) &= 275\end{aligned}$$

A positive covariance indicates that **X and Y tend to increase together**.

2. Correlation Coefficient

The **correlation coefficient** measures the strength and direction of the linear relationship between two variables.

The Pearson correlation coefficient is:

$$\begin{aligned}r &= \\ \frac{\text{Cov}(X,Y)}{\sigma_X \sigma_Y}\end{aligned}$$

Its value lies between:

$$-1 \leq r \leq 1$$

For these datasets:

$$r \approx 0.949$$

Interpretation

The correlation coefficient is approximately **0.949**, which is close to **+1**.

This indicates a **strong positive linear relationship** between `list_x` and `list_y`.

In simple terms:

As the values of X increase, the values of Y also tend to increase.

For example:

X: 10 → 20 → 30 → 40 → 50

Y: 15 → 25 → 35 → 45 → 60

Both datasets generally increase together.

Covariance vs Correlation

Feature	Covariance	Correlation
Purpose	Measures direction of joint variation	Measures strength and direction of linear relationship
Range	No fixed range	-1 to +1
Units	Has units of $X \times Y$	Unit-free
Positive value	Variables tend to increase together	Positive relationship

Negative value	Variables tend to move in opposite directions	Negative relationship
Interpretation	Depends on scale	Easier to interpret

Important Note

`np.cov()` uses **sample covariance** by default, so the program returns:

275

If **population covariance** is required, divide by (n) instead of (n-1):

1100

5

=220

So:

- **Sample covariance = 275**
- **Population covariance = 220**
- **Correlation coefficient ≈ 0.949**

The correlation coefficient remains the same regardless of whether sample or population covariance is used when calculated consistently.

Conclusion

For the given datasets:

```
list_x = [10, 20, 30, 40, 50]
```

```
list_y = [15, 25, 35, 45, 60]
```

the results are:

Covariance=275

Correlation≈0.949

The **positive covariance** shows that the variables move in the same direction, while the correlation coefficient of approximately **0.949** indicates a **strong positive linear relationship** between the two datasets.

Question 7: Draw a Boxplot and Identify Outliers

Given Data

```
data = [12, 14, 14, 15, 18, 19, 19, 21, 22, 22, 23, 23, 24, 26, 29, 35]
```

A **boxplot** is a statistical graph used to show the distribution of numerical data and identify potential **outliers**.

Python Program

```
import matplotlib.pyplot as plt
```

```
# Given data
```

```
data = [12, 14, 14, 15, 18, 19, 19, 21,

        22, 22, 23, 23, 24, 26, 29, 35]

# Draw boxplot

plt.boxplot(data)

plt.title("Boxplot of Given Data")

plt.ylabel("Values")

plt.show()

# Calculate Q1 and Q3 using the median-of-halves method

data_sorted = sorted(data)

n = len(data_sorted)

# Lower and upper halves

lower_half = data_sorted[:n // 2]

upper_half = data_sorted[n // 2:]
```

```
# Calculate Q1 and Q3

q1 = (lower_half[3] + lower_half[4]) / 2

q3 = (upper_half[3] + upper_half[4]) / 2


# Calculate IQR

iqr = q3 - q1


# Calculate lower and upper fences

lower_fence = q1 - 1.5 * iqr

upper_fence = q3 + 1.5 * iqr


# Identify outliers

outliers = [x for x in data if x < lower_fence or x > upper_fence]


print("Q1:", q1)

print("Q3:", q3)

print("IQR:", iqr)

print("Lower Fence:", lower_fence)

print("Upper Fence:", upper_fence)
```

```
print("Outliers:", outliers)
```

Expected Output

Q1: 16.5

Q3: 23.5

IQR: 7.0

Lower Fence: 6.0

Upper Fence: 34.0

Outliers: [35]

Step-by-Step Explanation

1. Sort the Data

The data is already sorted:

12, 14, 14, 15, 18, 19, 19, 21,

22, 22, 23, 23, 24, 26, 29, 35

There are **16 observations**.

2. Calculate Q1

The lower half contains:

12, 14, 14, 15, 18, 19, 19, 21

The middle two values are **15** and **18**.

Therefore:

$$\begin{aligned} Q1 &= \\ &= \frac{15 + 18}{2} \\ &= 16.5 \end{aligned}$$

So:

$$Q1 = 16.5$$

3. Calculate Q3

The upper half contains:

22, 22, 23, 23, 24, 26, 29, 35

The middle two values are **23** and **24**.

Therefore:

$$\begin{aligned} Q3 &= \\ \frac{23+24}{2} \\ &= 23.5 \end{aligned}$$

So:

$$Q3 = 23.5$$

4. Calculate IQR

The **Interquartile Range (IQR)** is:

$$\begin{aligned} \text{IQR} &= Q3 - Q1 \\ \text{IQR} &= 23.5 - 16.5 = 7 \end{aligned}$$

Therefore:

$$\text{IQR} = 7$$

5. Calculate Outlier Boundaries

The standard boxplot rule defines potential outliers as observations outside:

$$Q1 - 1.5(\text{IQR})$$

and

$$Q3 + 1.5(\text{IQR})$$

Lower Fence

$$16.5 - 1.5(7) \\ = 16.5 - 10.5 = 6$$

Upper Fence

$$23.5 + 1.5(7) \\ = 23.5 + 10.5 = 34$$

Therefore:

Lower Fence = 6
Upper Fence = 34

6. Identify the Outlier

Any value:

- Less than 6, or
- Greater than 34

is considered a potential outlier.

The dataset contains:

35

Since:

$$35 > 34$$

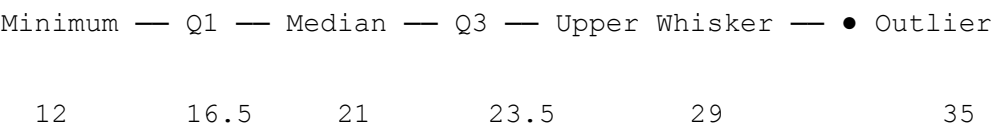
35 is an outlier.

Therefore:

Outlier=35

Interpretation of the Boxplot

The boxplot represents:



The value **35** appears separately from the main group of observations, indicating that it is a potential outlier.

Most of the observations are concentrated between approximately **16.5 and 23.5**, while 35 lies beyond the upper outlier boundary.

Important Boxplot Components

Component	Value
Q1	16.5
Median	21
Q3	23.5
IQR	7

Lower Fence	6
Upper Fence	34
Outlier	35

Why Are Boxplots Useful?

Boxplots are useful because they quickly show:

- Center of the data
- Spread of the data
- Median
- Quartiles
- Potential outliers
- Distribution and skewness

They are especially useful for comparing the distributions of multiple datasets.

Conclusion

The given dataset has:

$$Q1=16.5, \text{Median}=21, Q3=23.5$$

and:

$$IQR=7$$

The upper outlier boundary is:

$$Q3+1.5(IQR)=34$$

Since **35 > 34**, the value:

35

is identified as a **potential outlier**.

Thus, the boxplot shows that most values are concentrated in the central range, while **35 is unusually high compared with the rest of the dataset**.

Question 8: Relationship Between Advertising Spend and Daily Sales

Introduction

As a data analyst in an e-commerce company, **covariance and correlation** can be used to understand whether advertising spend and daily sales move together.

The two variables are:

```
advertising_spend = [200, 250, 300, 400, 500]
```

```
daily_sales = [2200, 2450, 2750, 3200, 4000]
```

Here:

- **Advertising spend** = amount spent on advertising each day.
 - **Daily sales** = sales generated on each day.
-

1. Using Covariance

Covariance measures the **direction of the relationship** between two variables.

Interpretation

- **Positive covariance** → Both variables tend to increase together.
- **Negative covariance** → One variable tends to increase when the other decreases.
- **Covariance near zero** → Little linear co-movement.

For this business problem, if covariance is positive, it suggests that higher advertising spending tends to occur with higher daily sales.

However, covariance depends on the units and scale of the variables, so its magnitude is difficult to compare across different datasets.

2. Using Correlation

Correlation measures both the **direction and strength of the linear relationship** between two variables.

The Pearson correlation coefficient ranges from:

$$-1 \leq r \leq 1$$

Interpretation

Correlation	Interpretation
$(r \approx +1)$	Strong positive relationship
$(r \approx 0)$	Little or no linear relationship

(r \approx -1) Strong negative relationship

Correlation is easier to interpret than covariance because it is **unit-free**.

Python Program

```
import numpy as np

# Given data

advertising_spend = [200, 250, 300, 400, 500]

daily_sales = [2200, 2450, 2750, 3200, 4000]

# Calculate covariance

covariance = np.cov(advertising_spend, daily_sales)[0, 1]

# Calculate correlation

correlation = np.corrcoef(advertising_spend, daily_sales)[0, 1]

# Display results

print("Covariance:", covariance)
```

```
print("Correlation Coefficient:", correlation)
```

Output

```
Covariance: 22500.0
```

```
Correlation Coefficient: 0.994...
```

Therefore:

Covariance=22500

Correlation≈0.994

Interpretation of the Result

The correlation coefficient is approximately:

$$r=0.994$$

This value is very close to **+1**.

Therefore, there is a **very strong positive linear relationship** between advertising spend and daily sales in this dataset.

In simple terms:

As advertising spend increases, daily sales also tend to increase.

For example:

Advertising Spend	Daily Sales
-------------------	-------------

200	→	2200
250	→	2450
300	→	2750
400	→	3200
500	→	4000

The data shows a clear upward pattern.

Business Interpretation

The marketing team can use this analysis as an initial indication that **higher advertising expenditure is associated with higher daily sales**.

For example:

- Spending ₹200 → Sales = ₹2,200
- Spending ₹300 → Sales = ₹2,750
- Spending ₹500 → Sales = ₹4,000

This suggests that advertising may be an important factor associated with sales performance.

However, **correlation does not prove causation**.

Other factors could also influence daily sales, such as:

- Discounts
- Seasonal demand
- Product availability

- Competitor pricing
- Holidays
- Website traffic
- Promotions
- Customer behavior

Therefore, the company should use additional analysis, such as **regression analysis or controlled experiments (A/B tests)**, before concluding that increasing advertising spend directly causes higher sales.

Covariance vs Correlation

Feature	Covariance	Correlation
Purpose	Measures direction of joint variation	Measures strength and direction of linear relationship
Value	Can range from negative to positive without a fixed bound	Between -1 and +1
Units	Has units	Unit-free
Interpretation	Direction is useful; magnitude depends on scale	Easier to interpret
Result here	22,500	≈ 0.994
Meaning	Positive relationship	Very strong positive relationship

Conclusion

Covariance and correlation are useful tools for exploring the relationship between **advertising spend and daily sales**.

For the given data:

Covariance=22500

Correlation≈0.994

The high positive correlation indicates a **very strong positive linear relationship**: higher advertising spend is associated with higher daily sales.

However, this result **does not establish that advertising spending causes the increase in sales**. Further analysis such as regression or experimentation would be needed to assess causality.

Question 9: Analyze Customer Satisfaction Survey Data

Given Data

```
survey_scores = [7, 8, 5, 9, 6, 7, 8, 9, 10, 4, 7, 6, 9, 8, 7]
```

The scores are measured on a scale of **1 to 10**. Before launching a new product, we should understand the **center, spread, and distribution** of customer satisfaction.

1. Summary Statistics to Use

A. Mean

The **mean** gives the average satisfaction score.

$$\text{Mean} = \frac{\text{Sum of scores}}{\text{Number of scores}}$$

For the given data:

$$\begin{aligned}\text{Mean} &= \\ &= \frac{115}{15} \\ &\approx 7.67\end{aligned}$$

Therefore, the average customer satisfaction score is approximately:

$$7.67/10$$

A high mean indicates generally positive customer satisfaction.

B. Median

The **median** is the middle value after arranging the scores in ascending order.

Sorted data:

4, 5, 6, 6, 7, 7, 7, 7, 8, 8, 8, 9, 9, 9, 10

There are 15 observations, so the 8th value is the median:

$$\text{Median} = 7$$

The median is useful because it is less affected by unusually high or low scores.

C. Mode

The **mode** is the most frequently occurring score.

In this dataset:

7 occurs 4 times

Therefore:

Mode=7

This tells us that a satisfaction score of **7 is the most common response**.

D. Standard Deviation

Standard deviation measures how spread out the satisfaction scores are around the mean.

A small standard deviation means that most customers give similar scores, while a large standard deviation indicates greater variation in customer opinions.

For this dataset, the population standard deviation is approximately:

1.59

This indicates a moderate amount of variation around the mean.

E. Minimum and Maximum

It is also useful to identify the minimum and maximum scores:

Minimum = 4

Maximum = 10

Therefore, the total range is:

$$\text{Range} = 10 - 4 = 6$$

This shows that customer responses range from relatively low satisfaction to very high satisfaction.

2. Visualizations to Use

A. Histogram

A **histogram** shows how frequently different score ranges occur.

It helps us identify:

- Concentration of scores
- Spread
- Shape of the distribution
- Possible unusual values

For satisfaction scores from 1 to 10, a histogram is a useful first visualization.

B. Boxplot

A **boxplot** can be used to identify:

- Median

- Quartiles
- Spread
- Potential outliers

It is useful for checking whether a few customers gave unusually low or high scores.

C. Bar Chart

Since satisfaction scores are discrete values from 1 to 10, a **bar chart of frequency by score** can also be useful for clearly showing how many customers selected each score.

3. Python Program to Create a Histogram

```
import matplotlib.pyplot as plt

# Survey data

survey_scores = [7, 8, 5, 9, 6, 7, 8, 9, 10,

                 4, 7, 6, 9, 8, 7]

# Create histogram

plt.hist(

    survey_scores,

    bins=range(1, 12),
```

```
        edgecolor="black"

    )

# Add labels and title

plt.title("Customer Satisfaction Score Distribution")

plt.xlabel("Satisfaction Score")

plt.ylabel("Number of Customers")


# Display the histogram

plt.show()
```

4. Calculate Summary Statistics Using Python

```
import statistics

survey_scores = [7, 8, 5, 9, 6, 7, 8, 9, 10,
                 4, 7, 6, 9, 8, 7]
```

```
mean = statistics.mean(survey_scores)

median = statistics.median(survey_scores)

mode = statistics.mode(survey_scores)

std_dev = statistics.pstdev(survey_scores)


print("Mean:", mean)

print("Median:", median)

print("Mode:", mode)

print("Standard Deviation:", std_dev)

print("Minimum:", min(survey_scores))

print("Maximum:", max(survey_scores))
```

Output

Mean: 7.666666666666667

Median: 7

Mode: 7

Standard Deviation: 1.5874507866387544

Minimum: 4

Maximum: 10

5. Interpretation of the Results

The results are:

Statistic	Value
Mean	7.67
Median	7
Mode	7
Standard Deviation	1.59
Minimum	4
Maximum	10
Range	6

The **mean of 7.67** indicates that overall customer satisfaction is fairly high.

The **median and mode are both 7**, showing that 7 is a central and common satisfaction score.

The **standard deviation of approximately 1.59** indicates that responses have some variation, but most scores are reasonably close to the average.

The histogram should show that most responses are concentrated around **7–9**, with fewer responses at the lower end such as 4 and 5.

6. Why This Analysis Is Important Before Product Launch

Understanding the distribution of satisfaction scores helps the marketing and product teams determine:

- Whether customers are generally satisfied.
- How consistent customer opinions are.
- Whether there are groups of dissatisfied customers.
- Whether extreme responses exist.
- Whether additional customer research is needed.

For example, although the average score is relatively high, the presence of scores such as **4 and 5** suggests that some customers may have significant concerns that should be investigated.

Conclusion

For customer satisfaction data, I would use:

- **Mean** → To measure average satisfaction.
- **Median** → To identify the central score.
- **Mode** → To identify the most common score.
- **Standard deviation** → To measure variation in responses.
- **Minimum/maximum and range** → To understand the overall spread.
- **Histogram** → To visualize the distribution of scores.
- **Boxplot** → To identify spread and potential outliers.

For the given data:

Mean≈7.67

Median=7

Mode=7

Standard Deviation $\approx$ 1.59

Overall, the survey suggests **generally positive customer satisfaction**, with most scores concentrated around 7–9.
